References

1 Kurtz, S. M. *et al.* Infection burden for hip and knee arthroplasty in the United States. *J Arthroplasty* **23**, 984-991 (2008). https://doi.org/10.1016/j.arth.2007.10.017

2 Premkumar, A. *et al.* Projected economic burden of periprosthetic joint infection of the hip and knee in the United States. *J Arthroplasty* **36**, 1484-1489.e1483 (2021). https://doi.org/10.1016/j.arth.2020.12.005

3 Koh, C. K. *et al.* Periprosthetic joint infection Is the main cause of failure for modern knee arthroplasty: An analysis of 11,134 knees. *Clin Orthop Relat Res* **475**, 2194-2201 (2017). https://doi.org/10.1007/s11999-017-5396-4

4 Esposito, C. I. CORR Insights(®): Periprosthetic joint infection Is the main cause of failure for modern knee arthroplasty: An Analysis of 11,134 Knees. *Clin Orthop Relat Res* **475**, 2202-2204 (2017). https://doi.org/10.1007/s11999-017-5436-0

5 Kurtz, S. M., Lau, E., Watson, H., Schmier, J. K. & Parvizi, J. Economic burden of periprosthetic joint infection in the United States. *Journal of Arthroplasty* **27**, 61-65.e61 (2012). https://doi.org/10.1016/j.arth.2012.02.022

6 Kapadia, B. H., Banerjee, S., Cherian, J. J., Bozic, K. J. & Mont, M. A. The economic impact of periprosthetic infections after total hip arthroplasty at a specialized tertiary-care center. *J Arthroplasty* **31**, 1422-1426 (2016). https://doi.org/10.1016/j.arth.2016.01.021

7 Kapadia, B. H. *et al.* The economic impact of periprosthetic infections following total knee arthroplasty at a specialized tertiary-care center. *J Arthroplasty* **29**, 929-932 (2014). https://doi.org/10.1016/j.arth.2013.09.017

8 Parisi, T. J., Konopka, J. F. & Bedair, H. S. What is the long-term economic societal effect of periprosthetic infections after THA? A Markov analysis. *Clin Orthop Relat Res* **475**, 1891-1900 (2017). https://doi.org/10.1007/s11999-017-5333-6

9 Sousa, A. *et al.* Economic impact of prosthetic joint infection - an evaluation within the Portuguese National Health System. *Journal of Bone and Joint Infection* **3**, 197-202 (2018). https://doi.org/10.7150/jbji.28508

10 Yao, J. J. *et al.* Direct inpatient medical costs of operative treatment of periprosthetic hip and knee infections are twofold higher than those of aseptic revisions. *Journal of Bone and Joint Surgery* **103**, 312-318 (2021). https://doi.org/10.2106/jbjs.20.00550

11 Patel, R. Periprosthetic joint infection. *New England Journal of Medicine* **388**, 251-262 (2023). https://doi.org/10.1056/NEJMra2203477

12 Hewlett, A. L., Kildow, B. J. & Cortés-Penfield, N. W. Periprosthetic joint infections. *Infect Dis Clin North Am* **39**, 399-417 (2025). https://doi.org/10.1016/j.idc.2025.02.010

13 Pivec, R., Johnson, A. J., Mears, S. C. & Mont, M. A. Hip arthroplasty. *The Lancet* **380**, 1768-1777 (2012). https://doi.org/https://doi.org/10.1016/S0140-6736(12)60607-2

14 Fisher, C. R. & Patel, R. Profiling the immune response to periprosthetic joint infection and non-infectious arthroplasty failure. *Antibiotics (Basel)* **12** (2023). https://doi.org/10.3390/antibiotics12020296

15 Schiffner, E. *et al.* Loosening of total knee arthroplasty - always aseptic? *J Clin Orthop Trauma* **11**, S234-s238 (2020). https://doi.org/10.1016/j.jcot.2019.05.001

16 Ledford, C. K., Perry, K. I., Hanssen, A. D. & Abdel, M. P. What are the contemporary etiologies for revision surgery and revision after primary, noncemented total hip arthroplasty? *J Am Acad Orthop Surg* **27**, 933-938 (2019). https://doi.org/10.5435/JAAOS-D-17-00842

17 Kenney, C., Dick, S., Lea, J., Liu, J. & Ebraheim, N. A. A systematic review of the causes of failure of revision total hip arthroplasty. *Journal of Orthopaedics* **16**, 393-395 (2019). https://doi.org/https://doi.org/10.1016/j.jor.2019.04.011

18 Mathis, D. T., Lohrer, L., Amsler, F. & Hirschmann, M. T. Reasons for failure in primary total knee arthroplasty - An analysis of prospectively collected registry data. *J Orthop* **23**, 60-66 (2021). https://doi.org/10.1016/j.jor.2020.12.008

19 Kelmer, G., Stone, A. H., Turcotte, J. & King, P. J. Reasons for revision: Primary total hip arthroplasty mechanisms of failure. *JAAOS - Journal of the American Academy of Orthopaedic Surgeons* **29**, 78-87 (2021). https://doi.org/10.5435/jaaos-d-19-00860

20 NIH consensus conference: Total hip replacement. NIH consensus development panel on total hip replacement. *Journal of the American Medical Association* **273**, 1950-1956 (1995).

21 Linke, S. *et al.* Microbiological pathogen analysis in native *versus* periprosthetic joint infections: A retrospective study. *J Orthop Surg Res* **17**, 9 (2022). https://doi.org/10.1186/s13018-021-02850-3

22 Azad, M. A. *et al.* Comparison of the BioFire Joint Infection Panel to 16S ribosomal RNA gene-based targeted metagenomic sequencing for testing synovial fluid from patients with knee arthroplasty failure. *J Clin Microbiol* **60**, e0112622 (2022). https://doi.org/10.1128/jcm.01126-22

23 Sabater-Martos, M., Clauss, M., Ribau, A., Sousa, R. & On Behalf Of The Leukocyte Count Synovial Fluid Working Group For The Unified Pji Definition Task, F. Differential synovial fluid white blood cell count for the diagnosis of chronic peri-prosthetic joint infection - a systematic review and meta-analysis. *J Bone Jt Infect* **10**, 165-184 (2025). https://doi.org/10.5194/jbji-10-165-2025

24 Nairn, L., Sivaratnam, S., Bali, K. & Wood, T. J. Neutrophil to lymphocyte ratio as an indicator of periprosthetic joint infection: A retrospective cohort study. *Journal of the American Academy Orthopaedic Surgeons* (2023). https://doi.org/10.5435/JAAOS-D-23-00600

25 Selsted, M. E., Harwig, S. S., Ganz, T., Schilling, J. W. & Lehrer, R. I. Primary structures of three human neutrophil defensins. *J Clin Invest* **76**, 1436-1439 (1985). https://doi.org/10.1172/jci112121

26 Ganz, T. Extracellular release of antimicrobial defensins by human polymorphonuclear leukocytes. *Infect Immun* **55**, 568-571 (1987). https://doi.org/10.1128/iai.55.3.568-571.1987

27 Deirmengian, C. *et al.* Validation of the alpha defensin lateral flow test for periprosthetic joint infection. *JBJS* **103** (2021).

28 Li, C. Y., Yam, L. T. & Lam, K. W. Esterases in human leukocytes. *Journal of Histochemistry & Cytochemistry* **21**, 1-12 (1973). https://doi.org/10.1177/21.1.1

29 Li, Z. *et al.* Alpha-defensin versus leukocyte esterase in periprosthetic joint infection: An updated meta-analysis. *Biomed Res Int* **2020**, 3704285 (2020). https://doi.org/10.1155/2020/3704285

30 Peng, X. *et al.* Synovial calprotectin for the diagnosis of periprosthetic joint infection: a diagnostic meta-analysis. *J Orthop Surg Res* **17**, 2 (2022). https://doi.org/10.1186/s13018-021-02746-2

31 Lee, Y. S. *et al.* Synovial fluid biomarkers for the diagnosis of periprosthetic joint infection: a systematic review and meta-analysis. *J Bone Joint Surg Am* **99**, 2077-2084 (2017). https://doi.org/10.2106/jbjs.17.00123

32 de Sandes Kimura, O. *et al.* Neutrophil extracellular trap-related biomarkers are increased in the synovial fluid of patients with periprosthetic joint infections. *Clinical Orthopaedics and Related Research* **482** (2024).

33 Lögters, T. *et al.* Diagnostic accuracy of neutrophil-derived circulating free DNA (cf-DNA/NETs) for septic arthritis. *J Orthop Res* **27**, 1401-1407 (2009). https://doi.org/10.1002/jor.20911

34 Parvizi, J. *et al.* New definition for periprosthetic joint infection: from the Workgroup of the Musculoskeletal Infection Society. *Clin Orthop Relat Res* **469**, 2992-2994 (2011). https://doi.org/10.1007/s11999-011-2102-9

35 Parvizi, J. & Gehrke, T. Definition of periprosthetic joint infection. *J Arthroplasty* **29**, 1331 (2014). https://doi.org/10.1016/j.arth.2014.03.009

36 Shohat, N. *et al.* Hip and knee section, what is the definition of a periprosthetic joint infection (PJI) of the knee and the hip? Can the same criteria be used for both joints? Proceedings of International Consensus on Orthopedic Infections. *J Arthroplasty* **34**, S325-s327 (2019). https://doi.org/10.1016/j.arth.2018.09.045

37 Trampuz, A. *et al.* Sonication of removed hip and knee prostheses for diagnosis of infection. *N Engl J Med* **357**, 654-663 (2007). https://doi.org/10.1056/NEJMoa061588

38 Berbari, E. F. *et al.* Culture-negative prosthetic joint infection. *Clin Infect Dis* **45**, 1113-1119 (2007). https://doi.org/10.1086/522184

39 Kalbian, I. *et al.* Culture-negative periprosthetic joint infection: Prevalence, aetiology, evaluation, recommendations, and treatment. *Int Orthop* **44**, 1255-1261 (2020). https://doi.org/10.1007/s00264-020-04627-5

40 Goswami, K. *et al.* An enhanced understanding of culture-negative periprosthetic joint infection with next-generation sequencing: A multicenter study. *J Bone Joint Surg Am* **104**, 1523-1529 (2022). https://doi.org/10.2106/JBJS.21.01061

41 Gomez, E. *et al.* Prosthetic joint infection diagnosis using broad-range PCR of biofilms dislodged from knee and hip arthroplasty surfaces using sonication. *J Clin Microbiol* **50**, 3501-3508 (2012). https://doi.org/10.1128/JCM.00834-12

42 Trampuz, A., Osmon, D. R., Hanssen, A. D., Steckelberg, J. M. & Patel, R. Molecular and antibiofilm approaches to prosthetic joint infection. *Clinical Orthopaedics and Related Research (1976-2007)* **414**, 69-88 (2003). https://doi.org/10.1097/01.blo.0000087324.60612.93

43 De Man, F. H. R. *et al.* Broad-range PCR in Selected episodes of prosthetic joint infection. *Infection* **37**, 292-294 (2009). https://doi.org/10.1007/s15010-008-8246-1

44 Vandercam, B. *et al.* Amplification-based DNA analysis in the diagnosis of prosthetic joint infection. *J Mol Diagn* **10**, 537-543 (2008). https://doi.org/10.2353/jmoldx.2008.070137

45 Panousis, K. *et al.* Poor predictive value of broad-range PCR for the detection of arthroplasty infection in 92 cases. *Acta Orthop* **76**, 341-346 (2005).

46 Fihman, V. *et al.* Improved diagnosis specificity in bone and joint infections using molecular techniques. *J Infect* **55**, 510-517 (2007). https://doi.org/10.1016/j.jinf.2007.09.001

47 Goswami, K., Parvizi, J. & Maxwell Courtney, P. Current recommendations for the diagnosis of acute and chronic PJI for hip and knee-cell counts, alpha-defensin, leukocyte esterase, next-generation sequencing. *Curr Rev Musculoskelet Med* **11**, 428-438 (2018). https://doi.org/10.1007/s12178-018-9513-0

48 Goswami, K. & Parvizi, J. Reinfection or persistence of periprosthetic joint infection? Next-generation sequencing reveals new findings. *Orthopaedic Proceedings* **103-B**, 34-34 (2021). https://doi.org/10.1302/1358-992x.2021.15.034

49 Roux-Dalvai, F. *et al.* Fast and accurate bacterial species identification in urine specimens using LC-MS/MS mass spectrometry and machine learning. *Mol Cell Proteomics* **18**, 2492-2505 (2019). https://doi.org/10.1074/mcp.TIR119.001559

50 Grenga, L., Pible, O. & Armengaud, J. Pathogen proteotyping: A rapidly developing application of mass spectrometry to address clinical concerns. *Clin Mass Spectrom* **14 Pt A**, 9-17 (2019). https://doi.org/10.1016/j.clinms.2019.04.004

51 O'Connor, K. *et al.* Proteomic profile at the time of surgery correlates with disease stage and surgical outcome in periprosthetic joint infection. *mBio* **16**, e0170025 (2025). https://doi.org/10.1128/mbio.01700-25

52 Alvarez Otero, J. *et al.* Evaluation of sonicate fluid culture cutoff points for periprosthetic joint infection diagnosis. *Open Forum Infect Dis* **11**, ofae159 (2024). https://doi.org/10.1093/ofid/ofae159

53 Franceschini, M., Pedretti, L., Cerbone, V. & Sandiford, N. A. Two stage revision: Indications, techniques and results. *Ann Jt* **7**, 4 (2022). https://doi.org/10.21037/aoj-20-84

54 Insall, J. N., Thompson, F. M. & Brause, B. D. Two-stage reimplantation for the salvage of infected total knee arthroplasty. *J Bone Joint Surg Am* **65**, 1087-1098 (1983).

55 Qasim, S. N., Swann, A. & Ashford, R. The DAIR (debridement, antibiotics and implant retention) procedure for infected total knee replacement - a literature review. *Sicot j* **3**, 2 (2017). https://doi.org/10.1051/sicotj/2016038

56 Pourani, Z. *et al.* Efficient periplasmic expression of active lysyl endopeptidase and optimizing the purification methods. *Protein Expression and Purification* **226**, 106618 (2025). https://doi.org/https://doi.org/10.1016/j.pep.2024.106618

57 Masaki, T., Fujihashi, T., Nakamura, K. & Soejima, M. Studies on a new proteolytic enzyme from *Achromobacter lyticus* M497-1. II. specificity and inhibition studies of Achromobacter protease I. *Biochim Biophys Acta* **660**, 51-55 (1981). https://doi.org/10.1016/0005-2744(81)90107-8

58 Masaki, T., Tanabe, M., Nakamura, K. & Soejima, M. Studies on a new proteolytic enzyme from *Achromobacter lyticus* M497-1. I. Purification and some enzymatic properties. *Biochim Biophys Acta* **660**, 44-50 (1981). ~~https://doi.org/10.1016/0005-2744(81)90106-6~~

59 Koga, H., Kaushik, S. & Cuervo, A. M. Protein homeostasis and aging: The importance of exquisite quality control. *Ageing Res Rev* **10**, 205-215 (2011). https://doi.org/10.1016/j.arr.2010.02.001

60 Sun, Z. & Brodsky, J. L. Protein quality control in the secretory pathway. *J Cell Biol* **218**, 3171-3187 (2019). https://doi.org/10.1083/jcb.201906047

61 Azam, T. A. & Ishihama, A. Twelve species of the nucleoid-associated protein from *Escherichia coli*: Sequence recognition specificity and dna binding affinity. *Journal of Biological Chemistry* **274**, 33105-33113 (1999). https://doi.org/https://doi.org/10.1074/jbc.274.46.33105

62 Stojkova, P., Spidlova, P. & Stulik, J. Nucleoid-associated protein HU: A Lilliputian in gene regulation of bacterial virulence. *Frontiers in Cellular and Infection Microbiology* **Volume 9 - 2019** (2019). https://doi.org/10.3389/fcimb.2019.00159

63 Harvey, K. L., Jarocki, V. M., Charles, I. G. & Djordjevic, S. P. The diverse functional roles of elongation factor Tu (EF-Tu) in microbial pathogenesis. *Front Microbiol* **10**, 2351 (2019). https://doi.org/10.3389/fmicb.2019.02351

64 Henderson, B. & Martin, A. Bacterial virulence in the moonlight: Multitasking bacterial moonlighting proteins are virulence determinants in infectious disease. *Infection and Immunity* **79**, 3476-3491 (2011). https://doi.org/10.1128/iai.00179-11

65 Smiline Girija, A. S. Moonlighting proteins [ML proteins]: The pandora's box of insidious oro-dental diseases. *Biochimica et Biophysica Acta (BBA) - Molecular Cell Research* **1870**, 119435 (2023). https://doi.org/https://doi.org/10.1016/j.bbamcr.2023.119435

66 Kainulainen, V. & Korhonen, T. K. Dancing to another tune-adhesive moonlighting proteins in bacteria. *Biology (Basel)* **3**, 178-204 (2014). https://doi.org/10.3390/biology3010178

67 Liu, D. & Bhunia, A. K. Anchorless bacterial moonlighting metabolic enzymes modulate the immune system and contribute to pathogenesis. *ACS Infectious Diseases* **10**, 2551-2566 (2024). https://doi.org/10.1021/acsinfecdis.4c00323

68 Hussain, M., Kohler, C. & Becker, K. Enolase of *Staphylococcus lugdunensis* is a surface-exposed moonlighting protein that binds to extracellular matrix and the plasminogen/plasmin system. *Frontiers in Microbiology* **Volume 13 - 2022** (2022). https://doi.org/10.3389/fmicb.2022.837297
69 Stojkova, P. & Spidlova, P. Bacterial nucleoid-associated protein HU as an extracellular player in host-pathogen interaction. *Front Cell Infect Microbiol* **12**, 999737 (2022). https://doi.org/10.3389/fcimb.2022.999737
70 Thakur, B., Arora, K., Gupta, A. & Guptasarma, P. The DNA-binding protein HU is a molecular glue that attaches bacteria to extracellular DNA in biofilms. *J Biol Chem* **296**, 100532 (2021). https://doi.org/10.1016/j.jbc.2021.100532
71 Widjaja, M. *et al.* Elongation factor Tu is a multifunctional and processed moonlighting protein. *Scientific Reports* **7**, 11227 (2017). https://doi.org/10.1038/s41598-017-10644-z
72 Zhang, Y., Yu, N. J. & Spremulli, L. L. Mutational analysis of the roles of residues in *Escherichia coli* elongation factor Ts in the interaction with elongation factor Tu. *J Biol Chem* **273**, 4556-4562 (1998). https://doi.org/10.1074/jbc.273.8.4556
73 Xu, B., Liu, L. & Song, G. Functions and regulation of translation elongation factors. *Front Mol Biosci* **8**, 816398 (2021). https://doi.org/10.3389/fmolb.2021.816398
74 Weijland, A., Harmark, K., Cool, R. H., Anborgh, P. H. & Parmeggiani, A. Elongation factor Tu: A molecular switch in protein biosynthesis. *Mol Microbiol* **6**, 683-688 (1992). https://doi.org/10.1111/j.1365-2958.1992.tb01516.x
75 Krab, I. M. & Parmeggiani, A. Mechanisms of EF-Tu, a pioneer GTPase. *Prog Nucleic Acid Res Mol Biol* **71**, 513-551 (2002). https://doi.org/10.1016/s0079-6603(02)71050-7
76 Furano, A. V. Content of elongation factor Tu in *Escherichia coli. Proc Natl Acad Sci U S A* **72**, 4780-4784 (1975). https://doi.org/10.1073/pnas.72.12.4780
77 Lazarev, V. F., Guzhova, I. V. & Margulis, B. A. Glyceraldehyde-3-phosphate dehydrogenase is a multifaceted therapeutic target. *Pharmaceutics* **12** (2020). https://doi.org/10.3390/pharmaceutics12050416
78 Mitsuzawa, H., Kimura, M., Kanda, E. & Ishihama, A. Glyceraldehyde-3-phosphate dehydrogenase and actin associate with RNA polymerase II and interact with its Rpb7 subunit. *FEBS Lett* **579**, 48-52 (2005). https://doi.org/10.1016/j.febslet.2004.11.045
79 Reed, G. H., Poyner, R. R., Larsen, T. M., Wedekind, J. E. & Rayment, I. Structural and mechanistic studies of enolase. *Curr Opin Struct Biol* **6**, 736-743 (1996). https://doi.org/10.1016/s0959-440x(96)80002-9
80 Jung, D. W. *et al.* A unique small molecule inhibitor of enolase clarifies its role in fundamental biological processes. *ACS Chem Biol* **8**, 1271-1282 (2013). https://doi.org/10.1021/cb300687k
81 Wery, M., Woldringh, C. L. & Rouviere-Yaniv, J. HU-GFP and DAPI co-localize on the *Escherichia coli* nucleoid. *Biochimie* **83**, 193-200 (2001). https://doi.org/10.1016/s0300-9084(01)01254-8
82 Kar, S., Edgar, R. & Adhya, S. Nucleoid remodeling by an altered HU protein: Reorganization of the transcription program. *Proc Natl Acad Sci U S A* **102**, 16397-16402 (2005). https://doi.org/10.1073/pnas.0508032102
83 Balandina, A., Kamashev, D. & Rouviere-Yaniv, J. The bacterial histone-like protein HU specifically recognizes similar structures in all nucleic acids. DNA, RNA, and their hybrids. *J Biol Chem* **277**, 27622-27628 (2002). https://doi.org/10.1074/jbc.M201978200
84 Dame, R. T. The role of nucleoid-associated proteins in the organization and compaction of bacterial chromatin. *Mol Microbiol* **56**, 858-870 (2005). https://doi.org/10.1111/j.1365-2958.2005.04598.x
85 Tan, T. L. *et al.* Culture-negative periprosthetic joint infection: An update on what to expect. *JBJS Open Access* **3**, e0060 (2018). https://doi.org/10.2106/JBJS.OA.17.00060